gently heated, it loses seven water molecules, and becomes anhydrous magnesium sulphate.

$$MgSO_4 \cdot 7H_2O \underset{\text{Cooling}}{\overset{\text{Heating}}{\rightleftharpoons}} MgSO_4 + 7H_2O$$

(Magnesium sulphate heptahydrate) (Anhydrous Magnesium sulphate)

If you add few drops of water or allow it to cool, the colourless anhydrous salt again turns back into hydrated salt.

## 9.7 Hygroscopy

Certain substances, when exposed to the atmospheric air at ordinary temperature, absorb moisture without changing their physical state. Such substances are called **hygroscopic substances** and this property is called hygroscopy.

Hygroscopic substances are used as drying agents.

**Example:** 1. Conc.Sulphuric acid ($H_2SO_4$).
2. Phosphorus Pentoxide ($P_2O_5$).
3. Quick lime (CaO).
4. Silica gel ($SiO_2$).

## 9.8 Deliquescence

Certain substances which are so hygroscopic, when exposed to the atmospheric air at ordinary temperatures, absorb enough water and get completely dissolved. Such substances are called **deliquescent substances** and this property is called **deliquescence**.

Deliquescent substances lose their crystalline shape and ultimately dissolve in the absorbed water forming a saturated solution.

Deliquescence is maximum when:

1) The temperature is low
2) The atmosphere is humid

**Examples:** Caustic soda (NaOH), Caustic potash (KOH) and Ferric chloride ($FeCl_3$).

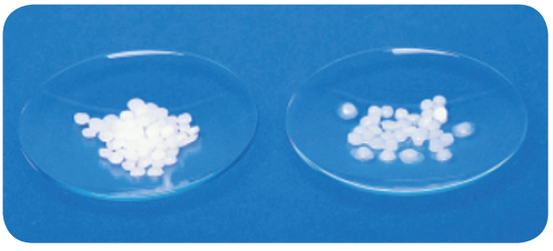


**Figure 9.11** Deliquescence in Sodium hydroxide

## 9.9 Problems Based on Solubility and Percentage by Mass and Volume

### I. Problems based on solubility

1) 1.5 g of solute is dissolved in 15 g of water to form a saturated solution at 298K. Find out the solubility of the solute at the temperature.

**Table 9.3** Difference between hygroscopic substances and deliquescence.

| Hygroscopic substances | Deliquescence substances |
|---|---|
| When exposed to the atmosphere at ordinary temperature, they absorb moisture and do not dissolve. | When exposed to the atmospheric air at ordinary temperature, they absorb moisture and dissolve. |
| Hygroscopic substances do not change its physical state on exposure to air. | Deliquescent substances change its physical state on exposure to air. |
| Hygroscopic substances may be amorphous solids or liquids. | Deliquescent substances are crystalline solids. |



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